

PRATHAMESH AKOLE

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🌐 [/Prathamesh](https://prathamesh.com)

🌐 [Portfolio](#)

Available from: May 2025

EDUCATION

Northeastern University - Khoury College of Computer Sciences

Master of Science in Computer Science

Sep 2023 – Present

GPA: 3.83 Boston, MA

Savitribai Phule Pune University

Bachelor of Technology in Information Technology

Aug 2019 – Jun 2023

Pune, MH, India

TECHNICAL SKILLS

Languages: Java, C, C++, Python, HTML, CSS, JavaScript

Frameworks: React.js, Node.js, Express.js, Spring Boot, JUnit, Selenium

Databases: NoSQL, MySQL, MongoDB, PostgreSQL, AWS DynamoDB

Cloud & DevOps: Amazon Web Services (AWS), Docker, Kubernetes, Kafka

Others: Linux, Shell, jQuery, JSON, RESTful APIs, Microservices, MVC, Git/GitHub, Jira, Agile Methodologies, Test-Driven Development

WORK EXPERIENCE

IDEaS Revenue Solutions

Jan 2023 – Jun 2023

Associate Software Developer Intern

Pune, MH

- Developed and executed new features using **Java, Spring** and wrote **RESTful APIs** for the flagship G2-RMS product.
- Engineered price forecasting features utilizing **Spring JPA, AWS DynamoDB**, and SQL queries, deploying on **AWS**.
- Implemented **Test Driven Development** using **JUnit, Mockito** resulting in reduction in post-release defects.
- Optimized complex **SQL** DB queries, reducing execution time and enhancing efficiency.

PowerMech Engineering Systems

Software Developer Engineer Intern

Pune, MH

- Engineered a **full-stack** rental platform with **ReactJS, NodeJS** and ML features, achieving 100% WCAG compliance.
- Implemented **CI/CD** pipelines reducing deployment time by 60% & increased test coverage to 95%.
- Wrote 50+ E2E integration tests in Cypress and 80+ Unit tests in JUnit for backend achieving 90% code coverage
- Leveraged **jQuery** to simplify DOM manipulation and improve UI/UX responsiveness.

PROJECTS

Distributed Key-Value Store with Fault Tolerance *Java, Distributed Systems, Docker*

- * Architected a distributed key-value store using **Java RMI**, and PAXOS consensus protocol achieving 99.9% data consistency across 5+ replicated nodes while maintaining fault tolerance with < 100ms latency
- * Containerized the app using **Docker** and automated deployment through **Bash** scripting, streamlining the workflow.
- * Engineered a fault-tolerant multi-threaded system achieving 100% availability with 3-second recovery.
- * Implemented **multithreading** in **Java** to handle concurrent client requests, improving system throughput by 50%

Pinpoint - A Geoguesser Social Media - *ReactJs, NodeJs, ExpressJs, Github*

- Architected and developed a full-stack application using **React.js, Node.js, ExpressJs with TypeScript**.
- Deployed and maintained scalable infrastructure using **Github Pages and Onrender**.
- Implemented secure **user authentication** and content moderation systems, focusing on accessibility and **UI/UX**
- Optimized performance with server-side caching, pagination, and efficient MongoDB data management techniques

Image Processing Application *Java, MVC, JUnit, SQL*

- Developed an **Image Processing** Application using **Design patterns**, following **SOLID** principles
- Attained 97% code coverage and minimized potential defects by rigorously writing **JUnit** tests for functionality testing
- Improved accessibility by integrating images with **SQL** using programming constructs like stored procedures & triggers

Airport Manager : An Airport Management Tool - *MySQL, React, Node*

- Designed a robust application based on database design principles using **MySQL**, ensuring optimal data retrieval
- Developed a user-accessible interface in **React**, allowing passengers to seamlessly access and view pertinent flight information. Utilized the **Node.js** web framework for backend to enhance system performance and ease API integration

TECHNICAL PUBLICATIONS

- Real-Time CCTV Violence Detection System Using Deep Learning, April 2023, ISBN: 978-1-6654-7524-2.
- Real-Time CNN-Based Face Mask Detection System, December 2022, ISBN: 9781032374598.